Random-walk level noise breaks anytime-valid trend monitoring: a failure theorem, a universal e-process and the price of robustness

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Abstract

Anytime-valid monitoring of a trend usually assumes that the noise around a smooth path is independent. We show that this fails under the smallest departure from it. If the latent level carries a random-walk disturbance of any positive variance, every fixed-shape contrast statistic has variance inflated by a factor that grows as the square of the series length, so that e-processes built on independent noise declare a change with probability tending to one; we give the exact rate and the sample size at which the failure begins. We then construct a universal e-process whose null class contains the random-walk level with unknown variance, an unknown scale of the noise, unknown regime levels and a linear trend, and prove that it is valid for every member of an arbitrary parametric Gaussian covariance class, whatever the mixture over alternatives. A decomposition separates the cost of robustness into the mixture over alternatives, the supremum over nuisance parameters and the information destroyed by the roughness itself, and a spike-in calibration gives the minimum detectable change. The motivating application is the Business Trends and Outlook Survey of the U.S. Census Bureau (56 waves on business use of artificial intelligence): the roughness is highly significant, an independence-based monitor declares changes in almost every series simulated from the fitted null, and the universal e-process finds no evidence of a change in growth, with a detectable change that is three to four times the average growth rate.

Keywords: anytime-valid inference; e-process; change detection; random-walk noise; spurious trend; survey time series.

1 Introduction

Users of a sequence of published estimates inspect each release and ask whether the growth of the indicator has changed. Testing at every release inflates error rates (Armitage et al., 1969), and the modern remedy is anytime-valid inference, in which e-values and test martingales control the error uniformly over time (Howard et al., 2021; Vovk and Wang, 2021; Ramdas et al., 2023; Grünwald et al., 2024). Constructions for change detection and sequential experiments take the observations to be, given the past, draws from a fixed parametric family with independent noise (Johari et al., 2022; Shin et al., 2023, 2024). For a smooth trend this means independent errors around a deterministic path, and the e-process monitors departures of the path.

The problem. Independent noise around a straight line is rarely credible for a published series. The latent level moves from wave to wave for reasons unrelated to growth, and a random walk with small variance is the simplest description (Harvey, 1990; Harvey and Chung, 2000; Durbin and Koopman, 2012). In the fixed-sample trend-testing literature a persistent noise component is

known to invalidate tests that assume independence, and it is the reason for spurious regressions (Granger and Newbold, 1974; Phillips, 1986) and for the robust trend tests of Vogelsang (1998), Bunzel and Vogelsang (2005), Harvey et al. (2007) and Perron and Yabu (2009). Those results do not transfer to the sequential setting. A fixed-sample test is wrong at one sample size; a monitor that is inspected at every release and uses a threshold that grows only logarithmically is wrong in a way that worsens with every release, and its error is not controlled by any rule that ignores the dependence. Nothing in the anytime-valid literature quantifies this, and none of its constructions has a null class that contains a rough level.

Contributions.

1. *A failure theorem* (Section 3). For any fixed-shape contrast of an equally spaced series, a random-walk level of variance p per wave and sampling variance σ^2 inflate the variance of the standardized statistic to $1 + (p/\sigma^2)\, c\, n^2\, (1 + o(1))$ with a constant $c > 0$ that depends on the contrast only (for curvature $c = 1/42$). The independence-based e-process therefore declares with probability tending to one, for every $p > 0$. The inflation is already 150-fold at the 54 waves of our application, and the failure begins when $n \approx \sigma/\sqrt{pc}$, about five waves for the curvature of the series studied here. The contrast form shows that the failure is not specific to the curvature statistic: it covers slope-change and level-shift contrasts.

2. *A universal e-process and its validity theorem* (Section 4). For a Gaussian null class whose covariance may be any consistent parametric family, we give an e-process that is the ratio of an arbitrary mixture over alternatives to the supremum of the invariant (restricted) likelihood of the null, and prove it valid for every member of the class (Theorem 2). The random-walk level, the first-order autoregressive level and a random slope are special cases. The class is correlated and has nuisance parameters in the covariance, which the existing universal-inference and invariance arguments do not cover.

3. *The price of robustness* (Sections 4–6). An exact inequality separates the loss into the mixture, the supremum and the information that the roughness destroys, a simulated decomposition shows that most of the loss is the mixture, and a spike-in calibration on the design of the application gives the minimum detectable change. On the 56 waves of the Business Trends and Outlook Survey (BTOS) on business use of artificial intelligence, the series could reveal a change of slope of 0.43 points per wave with probability 50 percent and 0.51 with probability 80 percent, against an average growth of 0.12.

The empirical finding is a null result and it is stated as one. The paper does not claim that the growth of business use of artificial intelligence is constant; it shows that this series cannot reveal a change smaller than three to four times its average growth once its roughness is accounted for, and that the changes that an independence-based monitor reports are produced by the roughness.

Related work. The e-process machinery we use is standard: Ville's inequality, mixture martingales (Robbins, 1970), invariance with respect to a location nuisance (Pérez-Ortiz et al., 2024) and the composite null construction of universal inference (Wasserman et al., 2020; Ruf et al., 2022). Sequential change detection is typically built on cumulative-sum or generalized likelihood-ratio statistics with calibrated thresholds (Page, 1954; Siegmund and Venkatraman, 1995; Lorden

and Pollak, 2005); e-detectors (Shin et al., 2024) give anytime-valid versions for independent observations. We do not propose a new mixture or a new detector. The contribution is the failure result, which has no counterpart in the anytime-valid literature, the validity theorem for a correlated null class, and the quantified cost. State-space models for survey series with a rough level (Scott and Smith, 1974; Durbin and Koopman, 2012; Harvey and Chung, 2000) estimate the trend and level; they do not give sequential error control, and their likelihood-ratio tests for roughness have non-standard null distributions (Self and Liang, 1987).

2 Setting

Let $y_1, \dots, y_n$ be published estimates at release times $t_1 < \cdots < t_n$ (in units of the release interval) with published design-based standard errors s_j, and let ρ_j label the regime (here, the questionnaire wording). Write $T_j = t_j - t_1$, $D = \mathrm{diag}(s_1^2, \dots, s_n^2)$ and $W_{jk} = \min(T_j, T_k)$. The null class is

$$y \sim \mathcal{N}\big(Xc,\ \kappa^2(D + r\,W)\big), \qquad c \in \mathbb{R}^q,\ \kappa^2 > 0,\ r \in \mathcal{R}_0, \tag{1}$$

where X has one column per regime (an unknown level for each) and one column t_j (an unknown common slope). Equation (1) is the marginal law of $y_j = L_j + e_j$ with sampling error $e_j \sim \mathcal{N}(0, \kappa^2 s_j^2)$ and level $L_j = L_1 + b\,T_j + \sum_{i \le j} \eta_i$, $\eta_i \sim \mathcal{N}(0, \kappa^2 r\,(T_i - T_{i-1}))$: a linear trend plus a random walk. The roughness $r \ge 0$ is measured relative to κ^2, and $p = \kappa^2 r$ is the variance of the level disturbance per wave. The roughness is a statistical description of the departures of the level from a line; we do not attribute it to a cause, and it is not identified against sampling error that is correlated across waves (for example through panel overlap), which the published standard errors, treating waves as independent, would understate. The set $\mathcal{R}_0$ is a finite grid that contains zero; we use 46 values spanning four orders of magnitude. The covariance of the first n waves is the leading block of that of the first $n+1$, so (1) defines a family of laws on the sequence.

Two families of alternatives are monitored, each at an unknown wave k and with an unknown size δ: a change of slope, $\mathbb{E}y = Xc + \delta\,(t - t_k)_+$, and a level shift, $\mathbb{E}y = Xc + \delta\,\mathbf{1}\{j \ge k\}$. A level shift of this kind is what an unannounced instrument event looks like. Both the horizon $n_{\max}$ and all grids and prior scales used below are fixed before the data are seen; the application is retrospective.

3 Independence-based monitors fail under level roughness

A contrast statistic. Take an equally spaced series, $t_j = j$, with $s_j = \sigma$. Let $a \in \mathbb{R}^n$ be a contrast orthogonal to the null design, $X^\top a = 0$ (for this section, $X = (1, j)$), and

$$Z_a = \frac{a^\top y}{\sigma \|a\|},$$

which is $\mathcal{N}(0,1)$ under independence. Curvature, a slope change at a fixed wave and a level shift at a fixed wave are all of this form after the null design is projected out. The independence-based monitor with prior scale τ declares when the mixture e-process $E_n(a) = (1 + \tau^2 B)^{-1/2} \exp\{\tau^2 A^2 / (2(1 + \tau^2 B))\}$, with $A = a^\top y / \sigma^2$ and $B = \|a\|^2 / \sigma^2$, reaches $2/\alpha$ (Robbins, 1970); the variant with a half-normal prior behaves identically for our purposes and is the one used in the simulations.

Table 1: Inflation of independence-based contrasts on the BTOS design (54 legacy-wording waves, fitted roughness $\hat{r} = 0.045$, scale $\hat{\kappa}^2 = 1.28$). λ: Rayleigh quotient $r\, a^\top W a / a^\top D a$; Var of Z: variance of the contrast standardized with the published standard errors, which equals $\hat{\kappa}^2(1+\lambda)$; the last column is the probability that a fixed-n test of nominal level 5% rejects.

Contrast	Inflation λ	Var of Z	False positives of a nominal 5% test (%)
curvature	151	194	89
slope change at wave 20	156	201	89
level shift at wave 20	72	94	84
slope change at wave 35	135	175	88
level shift at wave 35	97	125	86

Theorem 1 (failure)**.** *Let $y_j = \beta_0 + \beta_1 j + L_j + e_j$ with $e_j \sim \mathcal{N}(0, \sigma^2)$ independent and L a Gaussian random walk with increment variance $p > 0$. Let $a = a^{(n)}$ be the residual of $\psi(j/n)$ on $(1, j)$, where ψ is piecewise continuous on $[0,1]$ and is not almost everywhere equal to a linear function. Put $\tilde{\psi}$ for the residual of ψ on $(1,x)$ in $L_2[0,1]$ and $\Psi(x) = \int_x^1 \tilde{\psi}$. Then*

$$\mathrm{Var}(Z_a) = 1 + \lambda_n, \qquad \lambda_n = \frac{p}{\sigma^2}\,\frac{a^\top W a}{\|a\|^2} = \frac{p}{\sigma^2}\, c(\psi)\, n^2\,(1 + o(1)), \qquad c(\psi) = \frac{\int_0^1 \Psi^2}{\int_0^1 \tilde{\psi}^2} > 0.$$

Consequently, for every $\tau > 0$, $\alpha \in (0,1)$ and $p > 0$, the probability that $E_n(a) \geq 2/\alpha$ tends to one as $n \to \infty$. For curvature, $\psi(x) = x^2$ and $c = 1/42$.

The proof is in Appendix A. The quantity λ_n is the Rayleigh quotient of W against the sampling covariance on the contrast, and it is exact for any a; the asymptotic form gives the rate. Three consequences follow. First, the phase transition: the inflation exceeds one when $n > n^* = \sigma/\sqrt{p\,c}$; for the curvature of the BTOS design, with $r/s^2 \approx 2$, this is about five waves. Second, the statement is not an artefact of the half-normal mixture or of known scale; Proposition 1 treats estimation of the scale. Third, the failure is in the variance, not in a bias: the statistic is centred and a simple recalibration of its threshold would restore the level at one n but not uniformly over n, since the inflation grows.

Proposition 1 (unknown scale; proof sketch)**.** *For curvature, the same conclusion holds for the version of the e-process in which σ^2 is estimated from the residual sum of squares of the fit of $(1, j)$: the studentized statistic diverges at rate $\sqrt{n}$.*

Table 1 shows what this means for the application design: contrasts with the variance inflated 94-fold to 201-fold, and fixed-sample tests that reject in 84 to 89 percent of series simulated from a straight line. Figure 1 shows the anytime-valid case. With $p = 0.01$, under half the sampling variance $\sigma^2 = 0.0225$, the false-declaration probability of the monitor is 0.31 at 20 waves, 0.76 at 40, 0.97 at 80 and 1.0 at 160.

4 A universal e-process

4.1 Construction and validity

Let X_n be the design of the first n waves and R_n the data modulo the column space of X_n. Under (1) the law of R_n does not depend on c, and it has a density with respect to a fixed reference

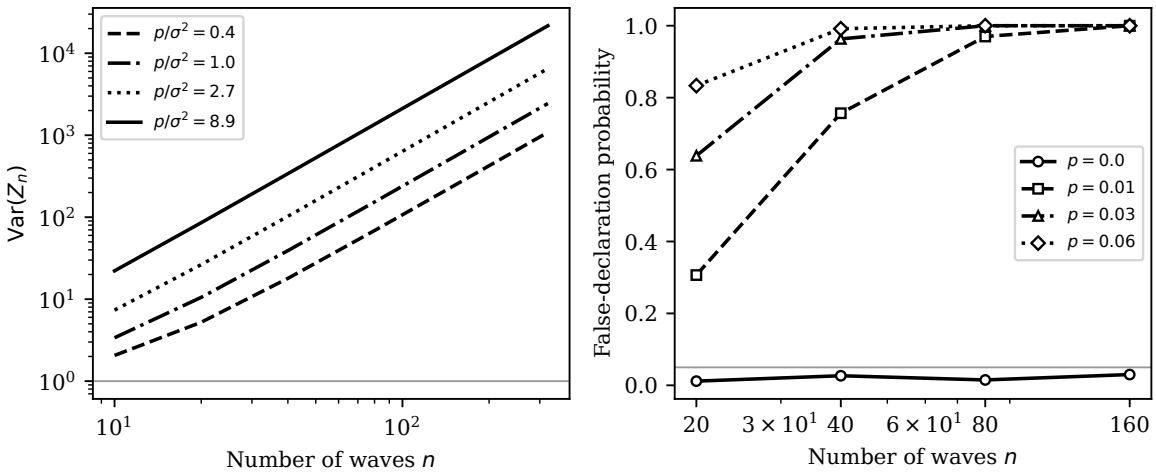


Figure 1: The independence-based e-process under a rough level. Left: probability of having declared a curvature change by wave n (known scale, threshold 40 on each of two one-sided monitors, level 0.05) for several p; sampling standard deviation 0.15. Right: the empirical variance of the standardized statistic against the theoretical $1 + (p/\sigma^2)n^2/42$.

measure on the residual space, of the restricted likelihood (REML) form

$$P_\theta(R_n) \propto |\Sigma_\theta|^{-1/2}\, |X_n^\top \Sigma_\theta^{-1} X_n|^{-1/2} \exp\big(-\tfrac{1}{2}\, y^\top P_\theta\, y\big), \qquad \Sigma_\theta = \kappa^2(D + rW),$$

with $P_\theta = \Sigma_\theta^{-1} - \Sigma_\theta^{-1} X_n (X_n^\top \Sigma_\theta^{-1} X_n)^{-1} X_n^\top \Sigma_\theta^{-1}$. A component $c = (g, f, k, \tau)$ of the alternative has a nuisance setting $g = (r_g, \kappa_g^2)$, a family f (slope or level), a change wave k and a prior scale τ for δ. With z the alternative column, $a = z^\top P_g y$ and $b = z^\top P_g z$ (with P_g computed at $\kappa^2 = 1$ scaled by κ_g^{-2}),

$$Q_c(R_n) = P_g(R_n) \cdot \mathrm{BF}_c, \qquad \mathrm{BF}_c = (1 + \tau^2 b)^{-1/2} \exp\Big(\frac{\tau^2 a^2}{2(1 + \tau^2 b)}\Big),$$

the density of R_n when $\delta \sim \mathcal{N}(0, \tau^2)$. Let $Q_n = \sum_c w_c Q_c$ with fixed weights and

$$E_n = \frac{Q_n(R_n)}{\sup_{\theta \in \Theta_0} P_\theta(R_n)}, \qquad \Theta_0 = \{(\kappa^2, r) : \kappa^2 > 0,\ r \in \mathcal{R}_0\}. \tag{2}$$

The supremum over κ^2 is explicit, $\hat\kappa^2 = y^\top P_{1,r}\, y/\nu$ with $\nu = n - q$, and the supremum over r is a search over the grid. The weights are uniform over the two families, the change waves $k \in \{3, \ldots, n_{\max}\}$, five roughness values from zero to 32 times the median sampling variance, four values of κ_g^2 from 0.5 to 4, and four prior scales per family (slope: 0.1, 0.2, 0.4, 0.8; level: 0.5, 1, 2, 4), about 8,300 components, a log-weight of 9 nats. Components whose change wave lies beyond the current wave have $z = 0$ on the data so far and contribute the null density.

Theorem 2 (validity)**.** *Let $\{\Sigma_\theta(n)\}_{\theta \in \Theta_0}$ be any collection of positive-definite covariance matrices, consistent in n (each $\Sigma_\theta(n)$ is the leading block of $\Sigma_\theta(n+1)$), and X_n designs for which the column space of X_{n+1} restricted to the first n rows is that of X_n. Let Q_n be any mixture of consistent Gaussian laws of R_n with positive-definite covariance, and E_n as in* (2)*. For every $\theta \in \Theta_0$ and c,*

if $y \sim \mathcal{N}(Xc, \Sigma_\theta)$ *then* $\mathbb{P}_\theta(\exists n : E_n \geq 1/\alpha) \leq \alpha$.

Corollary 1. *Theorem 2 covers the random-walk level of* (1)*, a first-order autoregressive level with unknown correlation, a random slope (integrated random walk), and any finite or countable union of such classes.*

The proof (Appendix A) shows that $\Lambda_n = Q_n/P_\theta$ is a likelihood-ratio martingale and that $E_n \leq \Lambda_n$ because the supremum is at least P_θ; Ville's inequality then gives the claim. The step that is not in the cited arguments is the first: the family of residual laws is consistent and the residual σ-fields are nested for a correlated null with nuisance parameters in the covariance, which is why the restricted likelihood, and not the likelihood, is the object. Nothing depends on how Q_n is mixed, so the construction is a template. The grid for $\mathcal{R}_0$ can be replaced by the continuum with the supremum in (2); validity holds for any set and the only requirement is that the supremum be attained or approximated from above by a measurable function.

4.2 The price of robustness

Proposition 2 (decomposition)**.** *For every component c with nuisance setting $g \in \Theta_0$,*

$$\log E_n \;\geq\; \underbrace{\log w_c}_{\textit{mixture}} + \underbrace{\log \Lambda_{c,n}}_{\textit{known-}\theta\textit{ LR}} - \underbrace{\Delta_n}_{\textit{supremum}}, \qquad \Delta_n = \log \sup_{\theta \in \Theta_0} P_\theta(R_n) - \log P_g(R_n) \;\geq 0, \tag{3}$$

where $\Lambda_{c,n} = Q_c/P_g$ is the likelihood ratio of a user who knows the nuisance setting g and the component.

The inequality is an identity-level bound and is loose in two ways. The mixture costs $\log(1/w_c)$, which is about 9 nats here. The supremum costs Δ_n, which under the null at g is a likelihood-ratio statistic for two parameters and is $O_P(1)$ for interior g; under an alternative the null fit can absorb signal, and Δ_n is not controlled by the inequality. Its size is therefore measured, not bounded. The first term of the known-θ benchmark grows linearly in information: for a simple alternative with slope change δ at wave k, $\mathbb{E} \log \Lambda = \frac{1}{2}\delta^2 b_k(\Sigma)$ with $b_k(\Sigma) = z_k^\top P\, z_k$, which is what roughness destroys. The information is a scale, not a detection threshold, and we calibrate detection by simulation rather than by a formula.

5 Simulation

All simulations use the dates and published standard errors of the 54 legacy-wording BTOS waves, a linear mean of slope 0.12 points per wave, level $\alpha = 0.05$ (threshold 20 for a single process) and $n_{\max} = 54$.

Validity. The probability that the universal e-process ever reaches 20 is 0.000 for independent errors, 0.000 for standard errors 1.5 times the published ones, and 0.000, 0.001 and 0.000 for random-walk levels with $p = 0.03$, 0.06 and 0.2 per wave (1,000 replications each), against 1.00 for the independence-based monitor at $p = 0.06$. For autoregressive levels with correlation 0.9 and 0.98, which lie outside the random-walk class, the universal e-process is also never above the threshold, and enlarging the class with an autoregressive-level setting changes the power at $p = 0.06$, slope 0.4

Table 2: Probability that the universal e-process declares by the last wave when the slope changes by δ points per wave, or the level shifts by δ points, at wave 25 of 54, by roughness p (variance of the level disturbance per wave; the published sampling variance is about 0.02). 500 replications.

	Slope change (points per wave)			Level shift (points)		
p	0.1	0.2	0.4	0.5	1	2
0	0.13	0.97	1.00	0.10	0.96	1.00
0.03	0.00	0.05	0.82	0.00	0.08	0.98
0.06	0.00	0.01	0.39	0.00	0.02	0.78

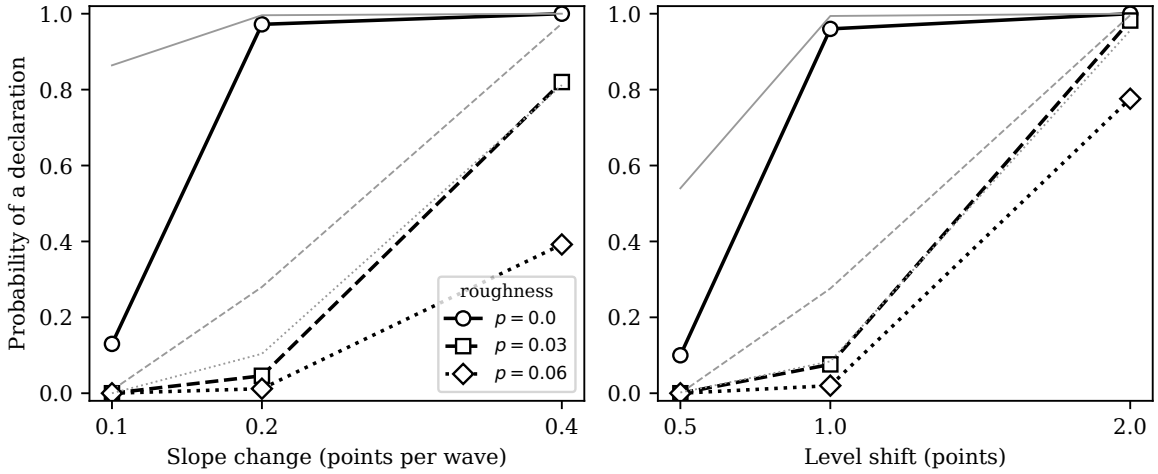


Figure 2: Power of the universal e-process for a slope change (left) and a level shift (right) at wave 25 of 54, by roughness p.

from 0.41 to 0.39 with no loss of validity. These probabilities are far below α: the construction is conservative, and Table 2 shows the price.

Power. Without roughness a slope change of 0.2 points per wave, nearly twice the typical growth, is detected in 97 percent of replications; with $p = 0.06$ the same change is detected in 1 percent, and detection needs a change of 0.4 (39 percent) or more (Table 2, Figure 2).

Where the cost comes from. Table 3 separates the three terms of (3). The known-θ benchmark, which knows the nuisance setting, the family, the size and the wave, detects a slope change of 0.4 at $p = 0.06$ in 99 percent of replications. Adding the supremum over the null parameters, for a single component, reduces this to 87 percent. The mixture over about 8,300 components reduces it to 40 percent. At smaller changes the supremum also costs a great deal (0.40 to 0.10 at slope 0.2), and the mixture takes the rest. The robustness is not mainly paid for by the supremum: it is paid for by the breadth of the alternatives.

Comparison with a calibrated test. A user who knew the null class could instead calibrate a likelihood-ratio monitor by simulation. We took the maximum over waves, families and change waves of the generalized likelihood ratio, with maximum-likelihood nuisance parameters, and set

Table 3: Detection probability by the last wave, at level 0.05, for the known-θ likelihood ratio, a single component with the supremum denominator, and the full universal e-process (400 replications, change at wave 25).

p	Alternative	Known-θ LR	One component, sup	Universal
0.06	slope change 0.4	0.99	0.87	0.40
0.06	slope change 0.2	0.40	0.10	0.01
0.06	level shift 2.0	0.99	0.95	0.77
0.06	level shift 1.0	0.21	0.07	0.02
0.03	slope change 0.2	0.76	0.32	0.04
0.03	level shift 1.0	0.51	0.27	0.09

Table 4: Detection probability of the calibrated generalized likelihood-ratio monitor and of the universal e-process (400 replications).

p	Alternative	Calibrated GLR	Universal e-process
0.0	slope change 0.2	1.00	0.99
0.06	slope change 0.2	0.04	0.01
0.06	slope change 0.4	0.23	0.39
0.03	slope change 0.2	0.10	0.03
0.0	level shift 1.0	0.99	0.96
0.06	level shift 1.0	0.07	0.03
0.06	level shift 2.0	0.70	0.78
0.03	level shift 1.0	0.12	0.10

the threshold at the 95 percent quantile of its null distribution at the least favourable roughness. Table 4 shows that this calibrated monitor is no more powerful than the universal e-process in the cases that matter (0.23 against 0.39 for slope 0.4 at $p = 0.06$; 0.70 against 0.78 for a level shift of 2), and is more powerful only where neither detects. The e-process buys exact anytime validity with no calibration and costs nothing in power relative to the calibrated alternative in this design.

6 Application: business use of artificial intelligence

Data. The BTOS is the Census Bureau's biweekly survey of about 1.2 million employer businesses (U.S. Census Bureau, 2025). We use the 56 national waves on current use of artificial intelligence from 24 September 2023 to 14 December 2025 (Bonney et al., 2024): 54 with the legacy wording and two with the wording introduced on 17 November 2025, which has its own level. The estimate rises from 3.7 to 10.0 percent under the legacy wording; the published standard errors have median 0.15.

Roughness and model choice. Maximizing the restricted likelihood over (1) gives $\hat{p} = 0.058$ points^2 per wave (95 percent profile interval 0.018 to 0.46) and $\hat{\kappa}^2 = 1.28$; the level disturbance has standard deviation 0.24 points per wave against a sampling standard deviation of 0.15. Setting $r = 0$ lowers the restricted log-likelihood by 39.8, a likelihood-ratio statistic of 79.6; because $r = 0$ is on the boundary, the reference distribution is the mixture $\frac{1}{2}\chi^2_0 + \frac{1}{2}\chi^2_1$ (Self and Liang, 1987) and the evidence is overwhelming. Table 5 compares models by AIC. A straight line with a random-walk

Table 5: Maximum-likelihood AIC of competing models for the legacy-wording national series on current use and on planned use of artificial intelligence (lower is better; regime levels are in all models).

Model (regime levels in all)	AIC, current use	AIC, planned use
line + independent noise	114.5	145.6
quadratic + independent noise	70.9	62.2
line + random-walk level	40.9	45.3
quadratic + random-walk level	41.9	42.0
line + random slope (integrated random walk)	52.5	53.9
line + random-walk level + random slope	42.9	47.1
line + AR(1) level (phi=0.9)	42.2	49.8

level is best; adding curvature does not improve it, and independent noise is far worse even with a quadratic mean. The data therefore support roughness and do not additionally support curvature.

What an independence-based monitor would say. The monitor of Section 3, restarted after each declaration, declares twice on the BTOS series, on 10 March 2024 (deceleration) and 6 April 2025 (acceleration). In 400 series simulated from the fitted null (straight line, estimated roughness and scale, BTOS dates and standard errors) it declares at least once in 100 percent and at least twice in 98 percent, 3.1 times on average. The two declarations are what the fitted null produces; they are not evidence of a change in growth.

The universal e-process. Applied to the 56 waves, the universal e-process peaks at $\log E = 0.39$ ($E = 1.5$) on 9 February 2025 against a threshold of $\log 20 = 3.0$ (Figure 3). By family, slope alternatives peak at -1.03 and level alternatives at 1.07, belonging to the one-wave fall of 10 March 2024. The result is not sensitive to the prior (Table 6). In the seven employment-size classes the roughness estimates range from 0.06 to 0.94 and no class exceeds $\log E = -0.6$; the average of the seven e-processes, valid under any dependence, peaks at -0.95. In a second series, the share of firms planning to use artificial intelligence, the roughness is $\hat{p} = 0.080$ (likelihood-ratio statistic 104.7) and the universal e-process peaks at -0.97.

A positive control. The method must be able to detect a real discontinuity. When the change of questionnaire wording on 17 November 2025 is treated as ordinary data, with no regime level, the universal e-process reaches $\log E = 17.2$ ($E \approx 2 \times 10^7$) on 30 November 2025. The monitor is therefore not insensitive; it simply does not declare changes that a rough level explains.

What the series could have revealed. We inject a change at wave 25 into series simulated from the fitted null on the full 56-wave design and record how often the universal e-process crosses 20 (300 replications per size; Figure 4). A slope change is detected with probability 50 percent at 0.43 points per wave and 80 percent at 0.51; a level shift at 1.75 and 2.12 points. The information scale of Proposition 2, $b_k \approx 205$ for a slope change (computed at $\kappa^2 = 1$; about 160 on the fitted scale), gives $\sqrt{2 \cdot 13/b_k}$ of 0.36 to 0.40, which is within 20 percent of the simulated threshold and shows that the information scale is a guide and not a substitute for the calibration. Without roughness

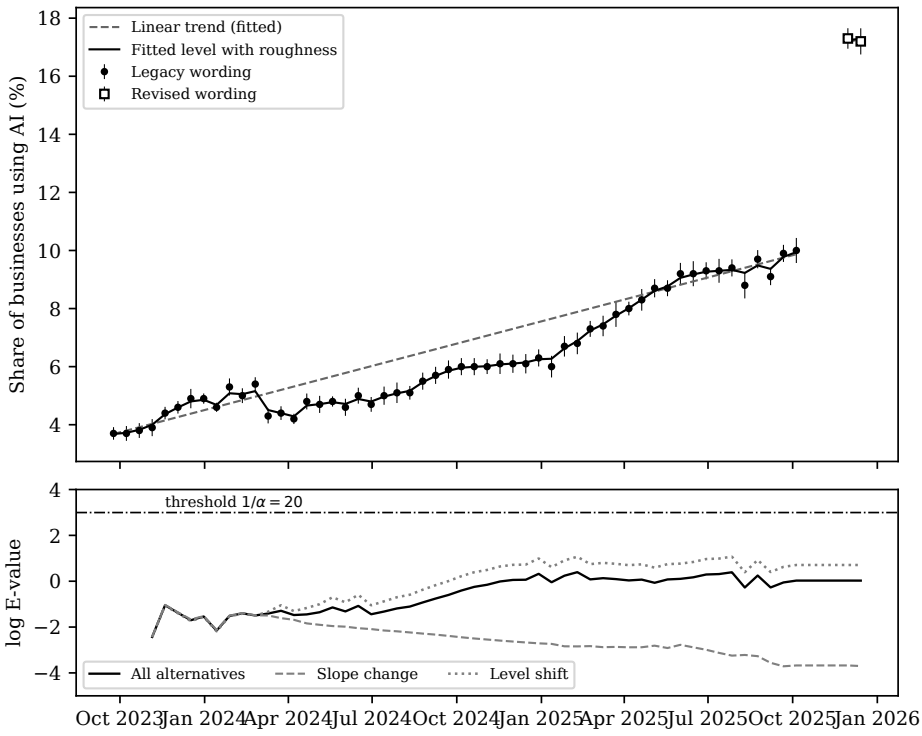


Figure 3: Top: national share of businesses using artificial intelligence with 95 percent intervals, the fitted linear trend and the fitted trend plus rough level. Bottom: log E-value of the universal e-process, for all alternatives and by family, against the threshold $\log 20$.

the information is 170 times larger. The largest change in the series, the one-wave fall of 1.1 points in March 2024, is below the minimum detectable level shift.

7 Discussion

What the paper shows. Anytime-valid monitoring of a trend is only as good as its null. If the latent level is rough, independence-based e-processes are not slightly anticonservative; they are wrong, by a margin that grows with the series, and the failure begins at roughness well below the sampling variance. A null class that includes the roughness restores validity at a price that we decompose and calibrate. In the application, the price is that the series cannot reveal a change smaller than three to four times its average growth, so the absence of declarations is not evidence of constant growth.

Limitations. The null class is Gaussian; the validity theorem covers any consistent parametric Gaussian covariance class but not non-Gaussian errors, and the failure theorem is for the benchmark monitor in an equally spaced design with a random-walk level. The universal construction is conservative, with realized false-declaration probabilities below 0.001; the calibrated alternative of Table 4 does not do better in this design but need not be exactly comparable elsewhere. The alternative mixture, the horizon and the grids are fixed in advance, and the application is retrospective. The standard errors enter as fixed weights, panel rotation is not modelled, and the roughness

Table 6: Maximum log E-value of the universal e-process on the BTOS national series under variants of the prior (threshold $\log 20 = 3.00$).

Variant	Maximum log E
Default	0.39
Wider prior scales on δ	0.10
Nuisance mixture restricted to rough settings ($r \geq 2\times$ sampling variance)	1.17
Scale fixed at the published value	1.25
Slope alternatives only	−1.03
Level alternatives only	1.07
Legacy-wording waves only	0.43

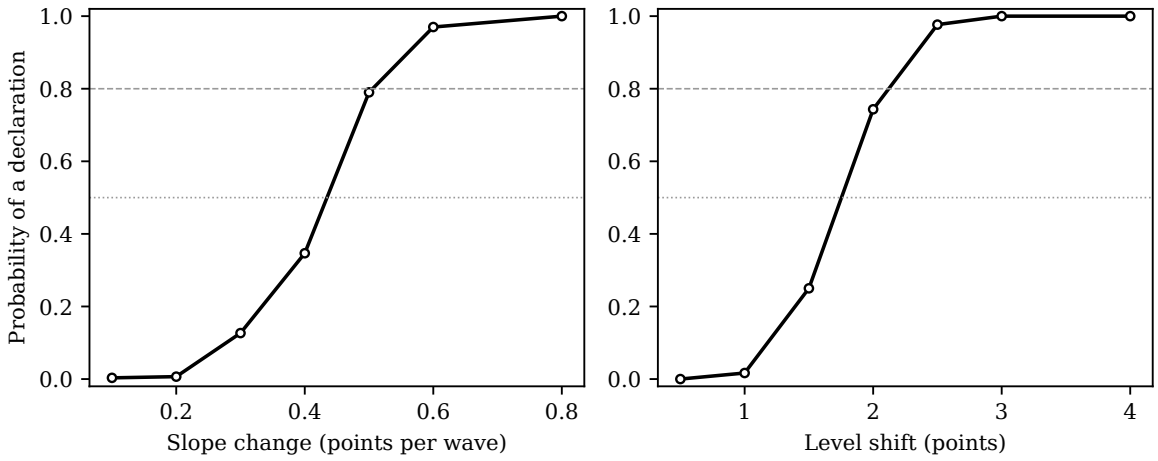


Figure 4: Probability that the universal e-process crosses 20 when a change is injected at wave 25 into series simulated from the fitted null on the BTOS design, for a slope change (left) and a level shift (right); the dotted lines mark 50 and 80 percent.

is not identified against correlated sampling error. We analyze two series from one survey; the roughness estimates in the size classes range from 0.06 to 0.94, and a replication in another survey is needed before the magnitude of the roughness is taken as general. We do not claim optimality: the cost decomposition measures the loss but no lower bound on it is proved. A tolerance for small departures from linearity, an exact-scale invariant version and a regret analysis are open.

Implications. A declaration from a monitor that assumes independent noise should be checked by estimating the roughness and simulating the monitor under the fitted null, as we did. For agencies, the information b_k and the spike-in curve are design quantities that say how large a change a series of a given length, precision and roughness can reveal, and they can be reported with the series.

Declaration of Generative AI Use

Generative AI tools were used to support the writing of code and to polish the language of the manuscript. The research idea, the methods and algorithms, and the interpretation of the results are the author's own. The author reviewed and edited all AI-assisted output and takes full responsibility

for the content of the work.

Data and code availability

All data are public statistics of the U.S. Census Bureau (Business Trends and Outlook Survey). The code reproduces every number, table and figure; the martingale and Ville-inequality steps of the validity proof are formalized in Lean 4 in the replication package.

References

Armitage, P., C. K. McPherson, and B. C. Rowe (1969). Repeated significance tests on accumulating data. *Journal of the Royal Statistical Society: Series A 132*(2), 235–244. https://doi.org/10.2307/2343787.

Bonney, K., C. Breaux, C. Buffington, E. Dinlersoz, L. Foster, N. Goldschlag, J. Haltiwanger, Z. Kroff, and K. Savage (2024). Tracking firm use of AI in real time: A snapshot from the Business Trends and Outlook Survey. NBER Working Paper 32319, National Bureau of Economic Research. https://doi.org/10.3386/w32319.

Bunzel, H. and T. J. Vogelsang (2005). Powerful trend function tests that are robust to strong serial correlation, with an application to the Prebisch–Singer hypothesis. *Journal of Business & Economic Statistics 23*, 381–394.

Durbin, J. and S. J. Koopman (2012). *Time Series Analysis by State Space Methods* (2nd ed.). Oxford: Oxford University Press. https://doi.org/10.1093/acprof:oso/9780199641178.001.0001.

Granger, C. W. J. and P. Newbold (1974). Spurious regressions in econometrics. *Journal of Econometrics 2*, 111–120.

Grünwald, P., R. de Heide, and W. M. Koolen (2024). Safe testing. *Journal of the Royal Statistical Society: Series B 86*(5), 1091–1128. https://doi.org/10.1093/jrsssb/qkae011.

Harvey, A. C. (1990). *Forecasting, Structural Time Series Models and the Kalman Filter*. Cambridge: Cambridge University Press. https://doi.org/10.1017/CBO9781107049994.

Harvey, A. C. and C.-H. Chung (2000). Estimating the underlying change in unemployment in the UK. *Journal of the Royal Statistical Society, Series A 163*.

Harvey, D. I., S. J. Leybourne, and A. M. R. Taylor (2007). A simple, robust and powerful test of the trend hypothesis. *Journal of Econometrics 141*, 1302–1330.

Howard, S. R., A. Ramdas, J. McAuliffe, and J. Sekhon (2021). Time-uniform, nonparametric, nonasymptotic confidence sequences. *Annals of Statistics 49*(2), 1055–1080. https://doi.org/10.1214/20-AOS1991.

Johari, R., P. Koomen, L. Pekelis, and D. Walsh (2022). Always valid inference: Continuous monitoring of A/B tests. *Operations Research 70*(3), 1806–1821. https://doi.org/10.1287/opre.2021.2135.

Lorden, G. and M. Pollak (2005). Nonanticipating estimation applied to sequential analysis and changepoint detection. *Annals of Statistics 33*, 1422–1454.

Page, E. S. (1954). Continuous inspection schemes. *Biometrika 41*(1–2), 100–115. https://doi.org/10.1093/biomet/41.1-2.100.

Pérez-Ortiz, M. F., T. Lardy, R. de Heide, and P. Grünwald (2024). E-statistics, group invariance and anytime-valid testing. *Annals of Statistics 52*(4). https://doi.org/10.1214/24-AOS2394.

Perron, P. and T. Yabu (2009). Estimating deterministic trends with an integrated or stationary noise component. *Journal of Econometrics 151*, 56–69.

Phillips, P. C. B. (1986). Understanding spurious regressions in econometrics. *Journal of Econometrics 33*, 311–340.

Ramdas, A., P. Grünwald, V. Vovk, and G. Shafer (2023). Game-theoretic statistics and safe anytime-valid inference. *Statistical Science 38*(4), 576–601. https://doi.org/10.1214/23-STS894.

Robbins, H. (1970). Statistical methods related to the law of the iterated logarithm. *Annals of Mathematical Statistics 41*(5), 1397–1409. https://doi.org/10.1214/aoms/1177696786.

Ruf, J., M. Larsson, W. M. Koolen, and A. Ramdas (2022). A composite generalization of Ville's martingale theorem. arXiv:2203.04485.

Scott, A. J. and T. M. F. Smith (1974). Analysis of repeated surveys using time series methods. *Journal of the American Statistical Association 69*(347), 674–678. https://doi.org/10.1080/01621459.1974.10480187.

Self, S. G. and K.-Y. Liang (1987). Asymptotic properties of maximum likelihood estimators and likelihood ratio tests under nonstandard conditions. *Journal of the American Statistical Association 82*, 605–610.

Shin, J., A. Ramdas, and A. Rinaldo (2023). E-detectors: A nonparametric framework for sequential change detection. *The New England Journal of Statistics in Data Science 2*(2), 229–260. https://doi.org/10.51387/23-NEJSDS51.

Shin, J., A. Ramdas, and A. Rinaldo (2024). E-detectors: a nonparametric framework for sequential change detection. *New England Journal of Statistics in Data Science 2*, 229–260.

Siegmund, D. and E. S. Venkatraman (1995). Using the generalized likelihood ratio statistic for sequential detection of a change-point. *Annals of Statistics 23*, 255–271.

U.S. Census Bureau (2025). Business Trends and Outlook Survey: Methodology. U.S. Census Bureau, Washington, DC. https://www2.census.gov/data/experimental-data-products/business-trends-and-outlook-survey/methodology.pdf.

Vogelsang, T. J. (1998). Trend function hypothesis testing in the presence of serial correlation. *Econometrica 66*, 123–148.

Vovk, V. and R. Wang (2021). E-values: Calibration, combination and applications. *Annals of Statistics 49*(3), 1736–1754. https://doi.org/10.1214/20-AOS2020.

Wasserman, L., A. Ramdas, and S. Balakrishnan (2020). Universal inference. *Proceedings of the National Academy of Sciences 117*(29), 16880–16890. https://doi.org/10.1073/pnas.1922664117.

A Proofs

Proof of Theorem 2. Fix $\theta \in \Theta_0$ and c. Let $\mathcal{G}_n$ be the σ-field generated by R_n. Since the column space of X_{n+1} restricted to the first n rows is that of X_n (a column for a new regime is zero on earlier rows), R_n is a function of R_{n+1} and $(\mathcal{G}_n)$ is a filtration. Under $\mathcal{N}(Xc, \Sigma_\theta)$ the law of R_n is $\mathcal{N}(0, M_n \Sigma_\theta M_n^\top)$ on the residual space, with density $P_\theta(R_n)$ with respect to a common reference measure. Each component of Q_n is a Gaussian density with positive-definite covariance, so $Q_n \ll P_\theta$ on $\mathcal{G}_n$ and $\Lambda_n = Q_n(R_n)/P_\theta(R_n)$. By consistency of the covariances and of the alternative columns, Q_{n+1} restricted to $\mathcal{G}_n$ is Q_n, and likewise P_θ; hence Λ_n is the Radon–Nikodym derivative of Q with respect to P_θ on $\mathcal{G}_n$, a nonnegative P_θ-martingale with $\Lambda_0 = 1$. Since $\theta \in \Theta_0$, $\sup_{\theta'} P_{\theta'}(R_n) \geq P_\theta(R_n)$, so $0 \leq E_n \leq \Lambda_n$. By Ville's inequality $\mathbb{P}_\theta(\sup_n \Lambda_n \geq 1/\alpha) \leq \alpha$, and $\{\exists n : E_n \geq 1/\alpha\}$ is contained in that event. The Corollary follows because each listed covariance family is consistent in n, and a union of classes is a larger Θ_0. □

Proof of Proposition 2. $E_n = Q_n / \sup_\theta P_\theta \geq w_c Q_c / \sup_\theta P_\theta = w_c \, \Lambda_{c,n} \, P_g / \sup_\theta P_\theta$. Take logarithms. □

Proof of Theorem 1. Write $S_i = \sum_{j \geq i} a_j$. Since $a \perp (1, j)$ the linear trend and L_1 do not enter $a^\top y = a^\top L + a^\top e$, and with $L_j = L_1 + \sum_{i=2}^{j} \eta_i$ we have $a^\top L = \sum_{i=2}^{n} S_i \eta_i$. By independence, $\mathrm{Var}(a^\top y) = p \sum_{i \geq 2} S_i^2 + \sigma^2 \|a\|^2$, and $\sum_{i \geq 2} S_i^2 = a^\top W a$ up to terms that vanish when $a \perp 1$. Hence $\mathrm{Var}(Z_a) = 1 + (p/\sigma^2)\, a^\top W a / \|a\|^2$. Since ψ is piecewise continuous, the residual $a_j = \tilde{\psi}(j/n) + O(n^{-1})$ uniformly off the $O(1)$ discontinuities, so $S_i = n\, \Psi(i/n) + O(1)$ and $\sum_i S_i^2 = n^3 \int_0^1 \Psi^2 \, (1 + o(1))$, while $\|a\|^2 = n \int_0^1 \tilde{\psi}^2 \, (1 + o(1))$. This gives the stated form. The constant $c(\psi)$ is positive: $\int \tilde{\psi}^2 > 0$ because ψ is not linear, and $\Psi \equiv 0$ would force $\tilde{\psi} = 0$ almost everywhere. For $\psi = x^2$, $\tilde{\psi} = (x - \frac{1}{2})^2 - \frac{1}{12}$, $\int \tilde{\psi}^2 = 1/180$ and $\Psi(x) = \int_x^1 \tilde{\psi}$ has $\int \Psi^2 = 1/7560$, so $c = 180/7560 = 1/42$.

For the e-process, Z_a is Gaussian with mean zero and variance $1 + \lambda_n \to \infty$, so $\mathbb{P}(|Z_a| > z_n) \to 1$ for any $z_n = o(\sqrt{\lambda_n})$. With $A^2/B = Z_a^2$, $\log E_n(a) = -\frac{1}{2} \log(1 + \tau^2 B) + \frac{1}{2} Z_a^2 \, \tau^2 B / (1 + \tau^2 B)$ and $B = \|a\|^2/\sigma^2 = O(n)$, which exceeds $\log(2/\alpha)$ once $Z_a^2 \geq 2 \log(2/\alpha) + \log(1 + \tau^2 B) + O(1) = O(\log n)$; this has probability tending to one since Z_a^2 is of order n^2. The half-normal mixture is bounded below by half the normal-mixture value times a constant factor, and the argument is unchanged. □

Sketch for Proposition 1. Let $\mathrm{RSS}_n^{(0)}$ be the residual sum of squares of y on $(1, j)$ and $\rho_n = (a^\top y)^2 / (\|a\|^2 \, \mathrm{RSS}_n^{(0)})$ with a the curvature contrast. The profile likelihood ratio of the quadratic model against the linear one is $(1 - \rho_n)^{-\nu/2}$ with $\nu = n - 2$, and a Laplace approximation of the mixture over the curvature gives $\log E_n^\sigma = \frac{\nu}{2} \log\{1/(1 - \rho_n)\} + O_P(\log n)$. By Donsker's theorem, $(np)^{-1/2} L_{\lfloor nx \rfloor}$ converges to a Brownian motion, and ρ_n converges in distribution to the squared cosine between the detrended Brownian path and the quadratic, which is positive almost surely. On

the event $\rho_n > \varepsilon$, whose probability tends to $\mathbb{P}(\rho_\infty > \varepsilon)$, $\log E_n^\sigma$ is of order n; letting $\varepsilon \to 0$ gives the claim. We present this as a sketch; the full argument requires uniform control of the Laplace remainder.

Computation. The universal e-process is computed on the log scale. For each wave n the matrices $P_{1,r} = \Sigma_1^{-1} - \Sigma_1^{-1} X (X^\top \Sigma_1^{-1} X)^{-1} X^\top \Sigma_1^{-1}$, $\Sigma_1 = D + rW$, are precomputed for the roughness grid, and the quadratic forms $y^\top P_{1,r} y$ and $z^\top P_{1,r} y$ give E_n in closed form.